$$\begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & -1 & -4 & 6 \\ 0 & 1 & 4 & -5 \end{bmatrix}$$

Row 3 = Row 3 + Row 2

$$\begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & -1 & -4 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The equations are: $x_1 + 3x_2 + 5x_3 = -2$, $-x_2 - 4x_3 = 6$ and $0 = 1$. This is a contradiction

$\boxed{\text{The equations are inconsistent}}$ □

13) $x_1 \quad - 3x_3 = 8$
$2x_1 + 2x_2 + 9x_3 = 7$
$x_2 + 5x_3 = -2$

Augmented Matrix: $\begin{bmatrix} 1 & 0 & -3 & 8 \\ 2 & 2 & 9 & 7 \\ 0 & 1 & 5 & -2 \end{bmatrix}$

Row 2 = Row 2 − 2 × Row 1 $\begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 1 & 5 & -2 \end{bmatrix}$

Row 3 = Row 3 − $\frac{1}{2}$ × Row 2

$$\begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 0 & -5/2 & 5/2 \end{bmatrix}$$

Row 3 = Row 3 × $\left(-\frac{2}{5}\right)$

$$\begin{bmatrix} 1 & 0 & -3 & 8 \\ 0 & 2 & 15 & -9 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

Row 1 = Row 1 + 3 × Row 3

$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 2 & 15 & -9 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

Row 2 = Row 2 − 15 × Row 3 → $\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 2 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{bmatrix}$

Row 2 = Row 2 × $\frac{1}{2}$

$$\begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$\boxed{x_1 = 5, x_2 = 3\ x_3 = -1}$

Determine if the systems in Exercises 15 and 16 are consistent.
Do not completely solve the systems.